# CIT 140 Analyst Portfolio Review

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Course: CIT 140

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Version: Final Portfolio Review

## Client Overview

This portfolio reviews my work from Weeks 1–7 in CIT 140. It gives the client a clear summary of the analysis completed, the evidence supporting each artifact, why the findings matter and what still needs to be checked. Each weekly case is kept separate and the original files remain in their original course folders.

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## Link to Repository README

CIT140-Week-1/README.md at main · laceymerced/CIT140-Week-1

## Week 1 – Problem Brief

**What the artifact shows:**

This problem brief explains the MR-17 incident, including the repeated retired test command, the effect on production and the information that still needed to be verified.

**Why it matters:**

It helped separate confirmed facts from assumptions and showed what evidence needed to be checked before deciding what caused the incident.

**What still needs to be checked:**

The source and ownership of legacy_sync.exe still needed confirmation, along with whether Test K-44 remained in another service or schedule.

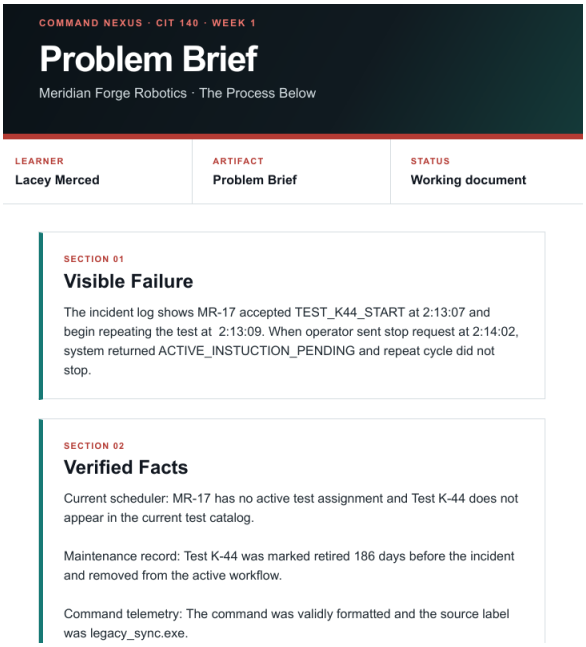



Figure 1. Week 1 Problem Brief excerpt. Source: Week 1 Problem Brief PDF.

[CIT140-Week-1/Merced_Lacey_CIT140_Week01_Problem_Brief.pdf at main · laceymerced/CIT140-Week-1]

## Week 2 – Business Case Recommendation

**What the artifact shows:**

This business case reviews whether the proposed work is worth considering and identifies important limits such as scope, time, resources, and available evidence.

**Why it matters:**

It helps determine whether the proposed work is realistic before time or resources are committed.

**What still needs to be checked:**

Any remaining assumptions about available resources, access, cost, or approval should be confirmed before moving forward.



MERIDIAN FORGE / ANALYST DOCUMENT · WEEK 02

Business Case Recommendation

QA Control Review Business Case Recommendation · Revision 3002

QA Control Review · A recommendation for the sponsor

1. Need, value and scope

**Proposed project**

Evaluate a bounded QA Control Review of the affected QA cell to examine command authorization, retired test controls and evidence preservation.

**Business need**

AQ and Engineering need a traceable record for further investigation because record retrieval required follow-up and ownership/source information was often incomplete.

**Expected value and beneficiaries**

QA and Engineering would benefit from a traceable record of command ownership, retired test controls and evidence preservation. Operations would also gain a stronger basis for deciding whether to request additional work but the review doesn't guarantee restored production.

**Included work**

Review command authorization ownership, scheduling and retirement controls and evidence preservation for the affected QA cell. Deliverables include a command ownership record, a scheduling/retirement control inventory and an evidence preservation checklist.

**Excluded work**

The review does not authorize restarting or operating MR-17, changing software or schedules, modifying original records, buying a replacement robot, selecting a vendor or extending the review to other cells.

Figure 2. Week 2 Business Case Recommendation excerpt. Source: Week 2 Business Case Recommendation PDF.

[CIT140-Week-1/Week 2/Merced_Lacey_CIT140_Week02_Business _Case.pdf at main · laceymerced/CIT140-Week-1]

## Week 3 – Stakeholder Map

**What the artifact shows:**

This stakeholder map identifies the people affected by the work and the people whose knowledge, input, or authority is needed.

**Why it matters:**

It helps show who should provide information, who may be affected by decisions, and who has responsibility or approval authority.

**What still needs to be checked:**

Any missing stakeholder roles, responsibilities, or approval limits should be confirmed before making recommendations.



MERIDIAN FORGE / WEEK 03 / ANALYST DOCUMENT

Stakeholder Map

Stakeholder Map · Revision 3324

Proposed QA Control Review / participation and decision rights

| PEOPLE | RELATIONSHIPS | NEXT CONVERSATIONS |
|---|---|---|
| Who is affected? | Who contributes and decides? | What needs confirmation? |

1. People and roles

**QA and night-shift operators**

Nia supervises night QA and can explain how the cell is used but a night shift operator should also participate because operators perform the checks and record exceptions. The operator has not yet been selected or interviewed, so that perspective still needs confirmation. (W03-S01)

**Engineering**

Elias can explain the command and scheduler records, review technical interpretations and advise on technical practicality. He needs read-only exports w/ source details and input from the record custodian. His expertise doesn't give him authority to approve spending or confirm the cause of the repeated command. (W03-S02)

**Operations**

Mara is helping coordinate the review and make sure the right people are included. She can help confirm contacts and organize the info but she isn't the perosn who can approve the budget or authorize the work. That still needs to to through the sponsor. (W03-S04)

**Sponsor and record custodian**

The sponsor can approve the review but we still need to confirm who that person is and what their

Figure 3. Week 3 Stakeholder Map excerpt. Source: Week 3 Stakeholder Map PDF.

[CIT140-Week-1/Week 3/Merced_Lacey_CIT140_Week03_Stakeholder_Map.pdf at main · laceymerced/CIT140-Week-1]

## Week 4 – Validated Requirements Set

**What the artifact shows:**

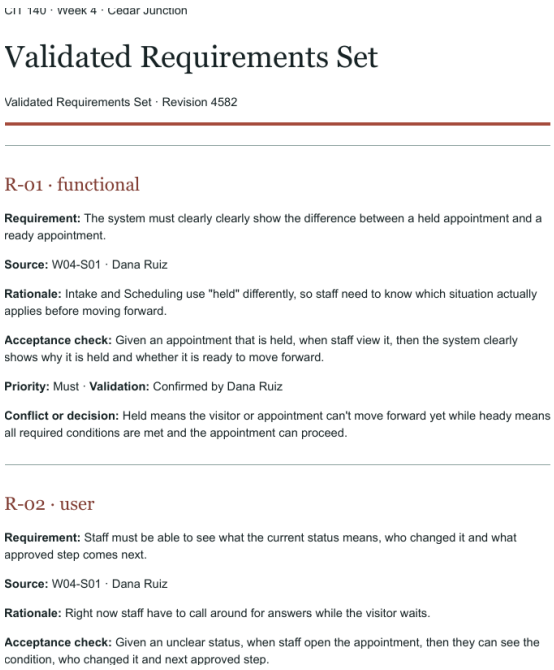
These requirements set turns stakeholder needs into clear requirements that can later be checked against a proposed solution.

**Why it matters:**

Clear requirements give the team a way to evaluate whether a future solution actually meets the identified needs.

**What still needs to be checked:**

The requirements should be reviewed again when solutions are compared to make sure each requirement can be tested and supported by evidence.



CIT 140 · Week 4 · Cedar Junction

Validated Requirements Set

Validated Requirements Set · Revision 4582

R-01 · functional

**Requirement:** The system must clearly clearly show the difference between a held appointment and a ready appointment.

**Source:** W04-S01 · Dana Ruiz

**Rationale:** Intake and Scheduling use "held" differently, so staff need to know which situation actually applies before moving forward.

**Acceptance check:** Given an appointment that is held, when staff view it, then the system clearly shows why it is held and whether it is ready to move forward.

**Priority:** Must · **Validation:** Confirmed by Dana Ruiz

**Conflict or decision:** Held means the visitor or appointment can't move forward yet while heady means all required conditions are met and the appointment can proceed.

R-02 · user

**Requirement:** Staff must be able to see what the current status means, who changed it and what approved step comes next.

**Source:** W04-S01 · Dana Ruiz

**Rationale:** Right now staff have to call around for answers while the visitor waits.

**Acceptance check:** Given an unclear status, when staff open the appointment, then they can see the condition, who changed it and next approved step.

Figure 4. Week 4 Validated Requirements Set excerpt. Source: Week 4 Requirements PDF.

[CIT140-Week-1/Week 4/Merced_Lacey_CIT140_Week04_Validated_Requirements_Set.pdf at main · laceymerced/CIT140-Week-1]

## Week 5 – As-Is Process Model

**What the artifact shows:**

This process model shows how the current work happens, including the major activities, decisions, and handoffs between people or roles.

**Why it matters:**

It makes the current process easier to understand and helps identify where delays, confusion, or missing information may occur.

**What still needs to be checked:**

Any unclear handoff or decision point should be confirmed with the appropriate source before recommending a future process.

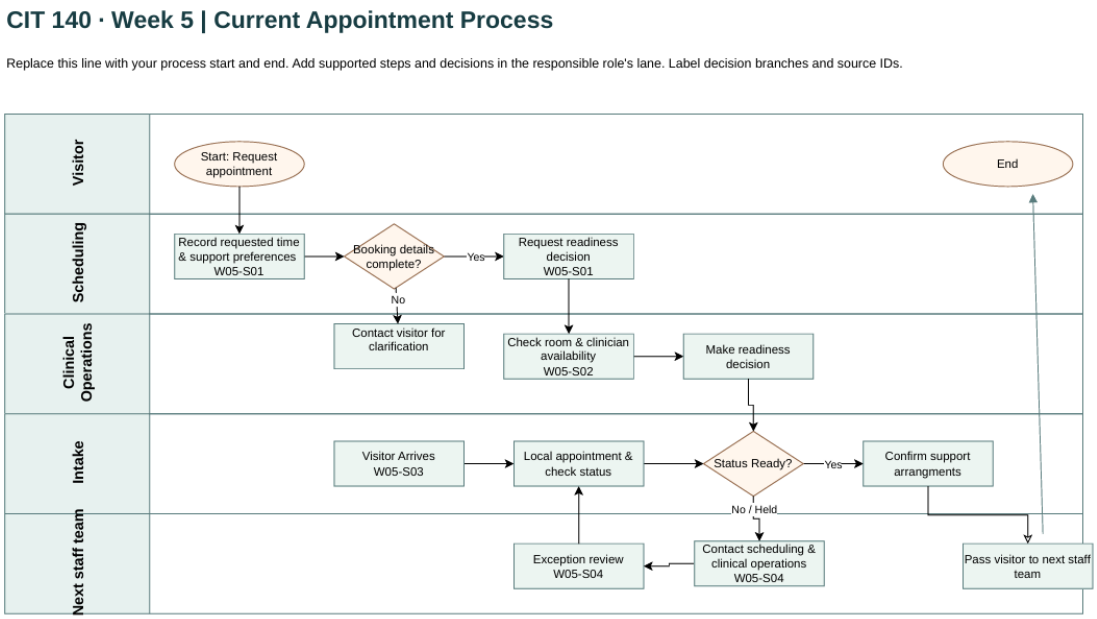



Figure 5. Week 5 As-Is Process Model excerpt. Source: Week 5 Process Model PDF.

[CIT140-Week-1/Week 5/Artifact/Merced_Lacey_CIT140_Week05_As_Is_Pricess_Model.drawio.pdf at main · laceymerced/CIT140-Week-1]

## Week 6 – Data-Flow Model

**What the artifact shows:**

This data-flow model shows how information moves between people, processes, systems, and records.

**Why it matters:**

It helps identify where information comes from, where it goes, and where missing or unclear data could affect the process.

**What still needs to be checked:**

Any unclear data source, ownership, or handoff should be confirmed before making a design decision.

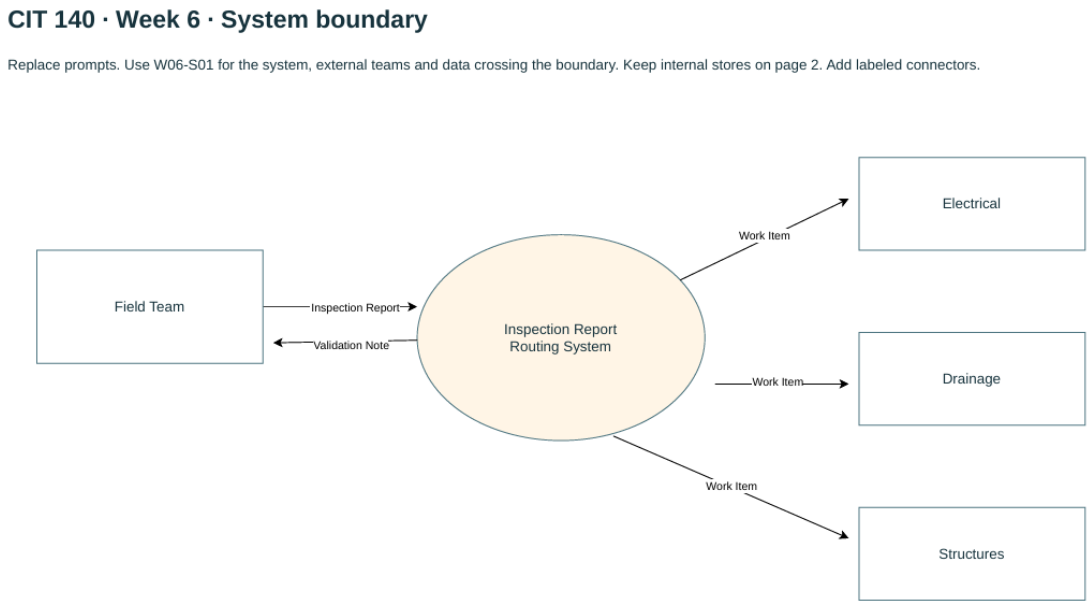



Figure 6. Week 6 Data-Flow Model excerpt. Source: Week 6 Data-Flow Model PDF.

CIT140-Week-1/Week06/Artifact/Merced_Lacey_CIT140_Week06.pdf at main · laceymerced/CIT140-Week-1

# Week 7 – Data-Quality Report

**What the artifact shows:**

This data-quality report checks 20 inspection records using rules for required values, repeated work orders, valid status values, inspection duration, and inspector information.

**Why it matters:**

It identifies records that may not be reliable enough for their intended use and shows which records require follow-up instead of guessing at missing information.

**What still needs to be checked:**

The repeated Work_Order_ID records should be verified, and the missing or out-of-range inspection durations for R10, R18, and R20 should be checked against the original inspection records.

**CIT 140 · Week 7 · Data-Quality Report**

Meridian Forge · fictional inspection export v1

| | |
|---|---|
| **Learner name / date** | Lacey Merced 11/2/2026 |
| **Import status** | 20 rows loaded |

Amber cells: your responses. Other results calculate from Prepared and Audit.

| Rule | Check | Rows flagged | Share of all rows | Affected Row_IDs |
|---|---|---|---|---|
| Q01 | Machine ID is required | 2 | 10% | R03, R14 |
| Q02 | Work order is unique | 2 | 10% | R05, R06 |
| Q03 | Status uses the agreed list | 2 | 10% | R08, R16 |
| Q04 | Duration is in range | 3 | 15% | R10, R18, R20 |
| Q05 | READY identifies an inspector | 3 | 15% | R03. R07, R12 |
| **Distinct rows with any issue** | | 11 | 55% | Count each row once, even when several Q-rules flag it. Do not add the five counts. |

Figure 7. Week 7 Data-Quality Report excerpt. Source: Week 7 Data-Quality Report.

[CIT140-Week-1/Week07/Artifact/Merced_Lacey_CIT140_Week07 (Revised 11.2.26).pdf at main · laceymerced/CIT140-Week-1]

## Closing Review

Across Weeks 1–7, my analysis moved from identifying problems to reviewing whether further work was justified, identifying stakeholders, defining requirements, mapping processes and information flow, and checking data quality. The work shows how evidence can be used to support findings instead of relying on assumptions.

Some questions still require confirmation from original records or responsible stakeholders. Before comparing solutions or designing a future system, those remaining gaps should be reviewed so that decisions are based on reliable evidence.

# Evidence List

Week 1: Problem Brief – original Week 1 Problem Brief PDF

Week 2: Business Case Recommendation – original Week 2 recommendation PDF

Week 3: Stakeholder Map – original Week 3 map/report PDF

Week 4: Validated Requirements Set – original Week 4 requirements PDF

Week 5: As-Is Process Model – original Week 5 editable diagram and PDF

Week 6: Data-Flow Model – original Week 6 editable diagram and PDF

Week 7: Data-Quality Report – original Week 7 Excel workbook, Report PDF, and Applied Steps screenshot